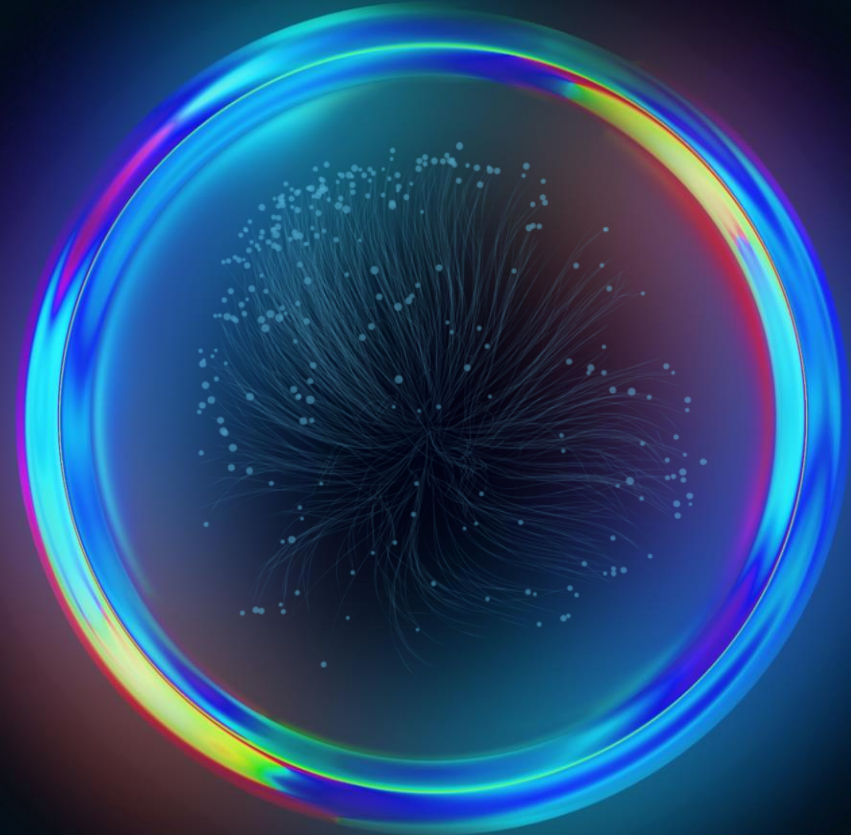


The evolving future of AI and Generative AI

Perspective

November 2024



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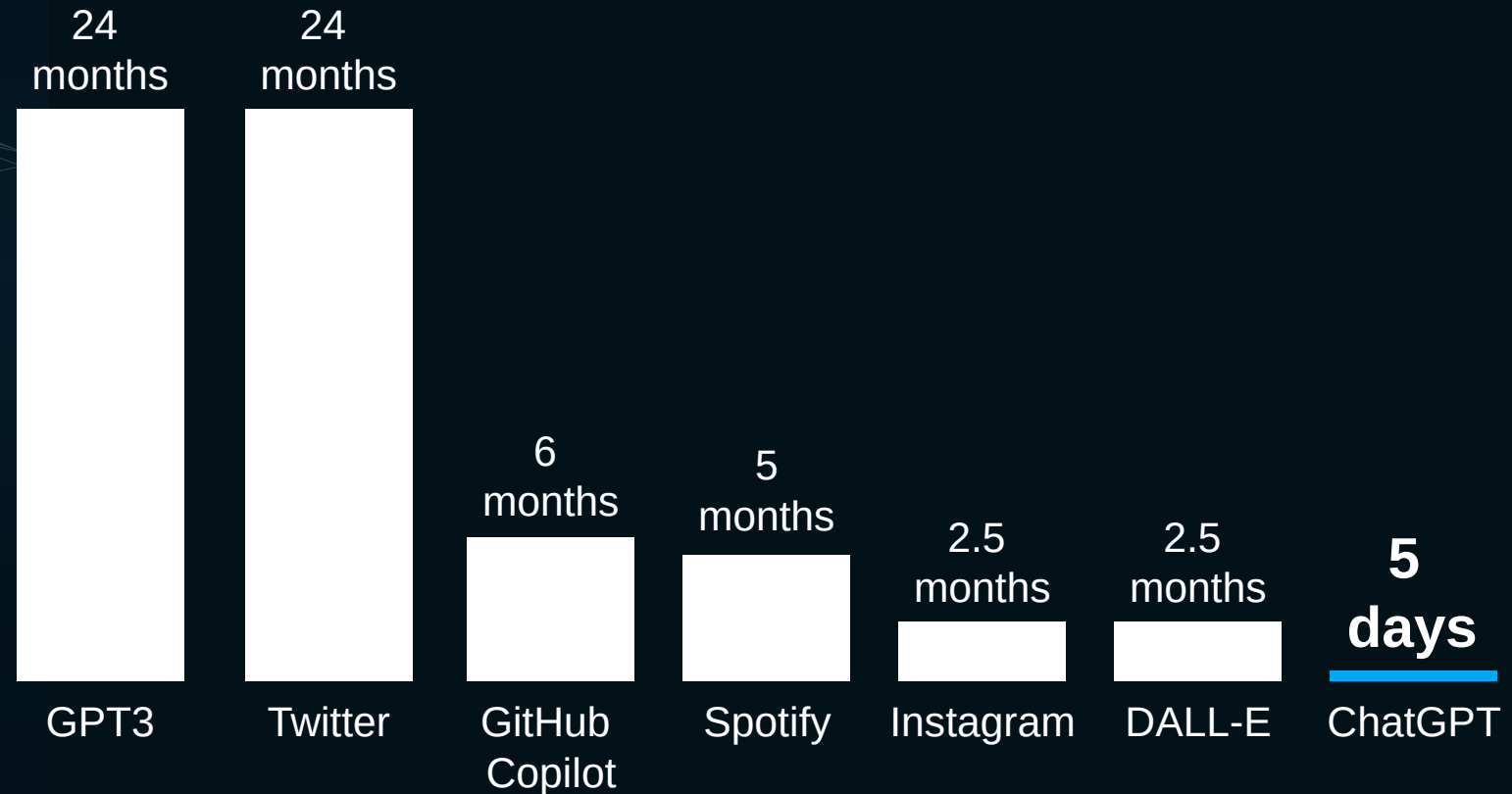
Agenda

The potential of Generative AI

Demos – Virtual Agents and LegacyX

Adoption rates are reaching scale faster than any previous commercial platform

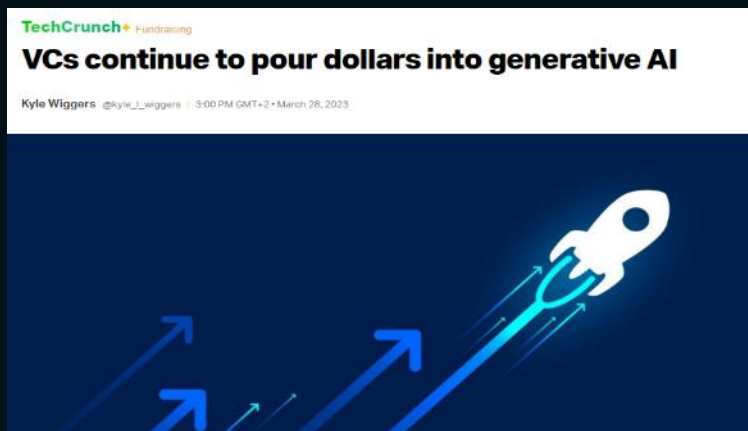
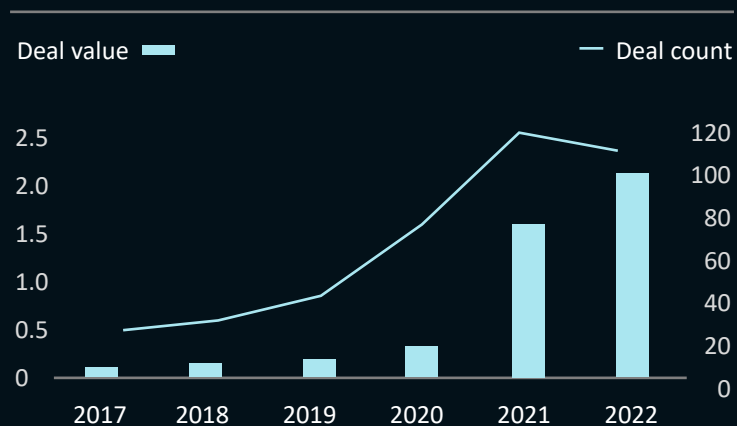
Time to 1 million users¹



Early & rapid adoption of Generative AI indicates that the impact in day-to-day life is taking off

1. OpenAI and statista data

We see surging investment levels in Generative AI



425%

Increase of VC investments
in Generative AI since 2020

>\$8Bn

Invested into GenAI startups
currently and rapidly growing

+\$10Bn

Valuation by Microsoft in
generative AI leader OpenAI

GenAI is powered by Foundation Models that can be adapted to a wide range of tasks

Language model example (GPT-3)

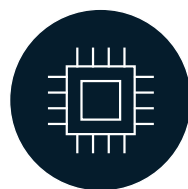


Large scale input datasets...

45TB data¹

For example, for GPT-3:

- Crawl of the internet
- Reddit-linked content
- >250K Books
- Wikipedia



are fed into a Transformer Model...

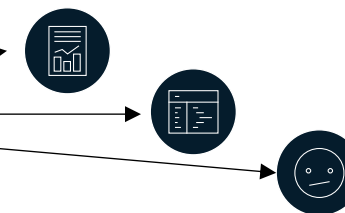
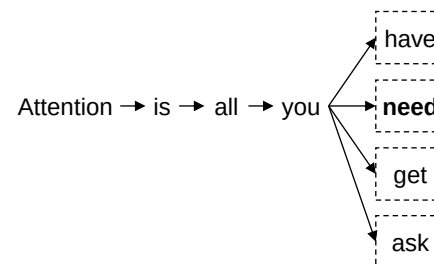
175B parameters



that is trained on a single task...

32yrs training on 1 GPU²

For example, predict next word



but yields a Foundation Model that generalizes to many text-generation tasks

800GB model size

- Write articles
- Write code
- Interpret emotions in text

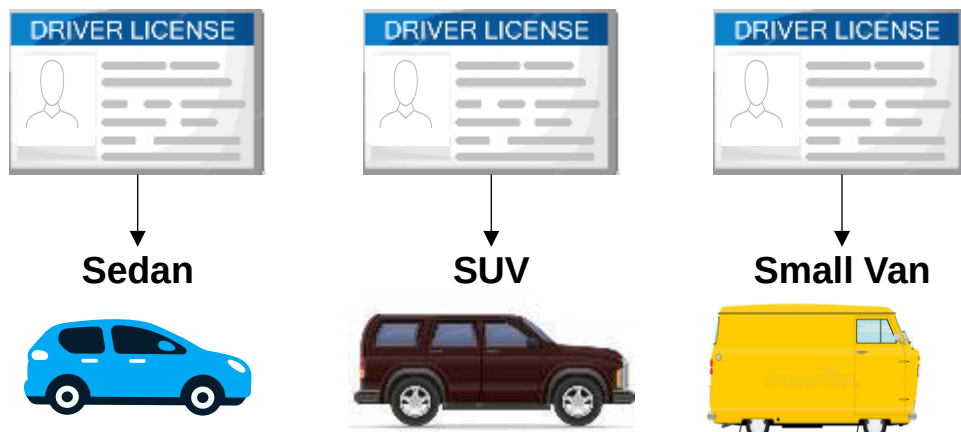
1. Compressed text from common crawl datasets

2. Using currently fastest GPU (Nvidia A100, costing ~\$30K), actual training is done on 100-1000s of units in parallel

What is the paradigm shift with foundation models? An analogy...

Traditional ML techniques

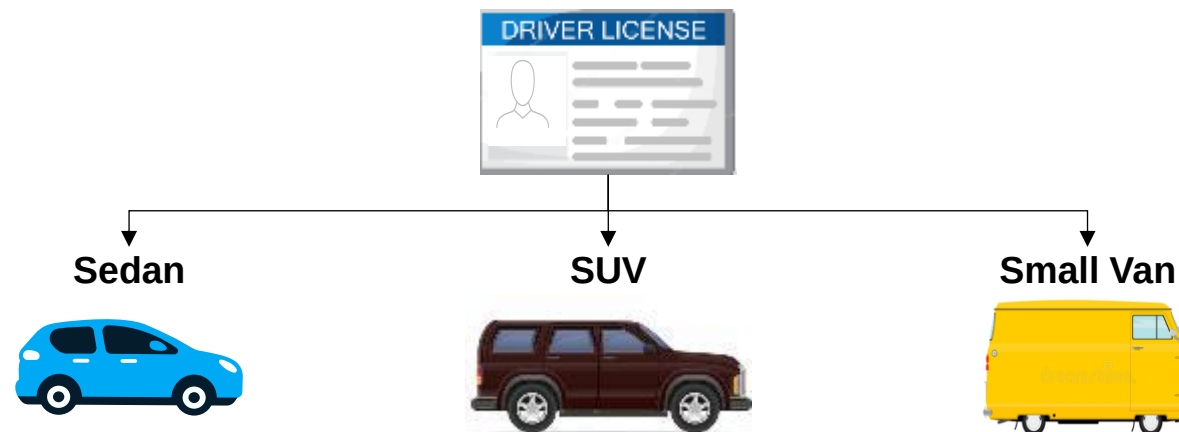
Custom-made solutions for each application



- AI trained with supervision, using specific data, on a single task
- Model needs to be **retrained for each new use case**, as needing to re-learn how to drive (and get a license) every time you switch to a new vehicle

Foundation models¹

One core pre-trained model powering many end applications



- AI self trained, using broad data, to perform multiple tasks and be finetuned to learn more tasks
- With Foundation Models, the **model has learned the “representation” of the use case** (drive a 4-wheel vehicle), and can apply to multiple use-cases (drive new vehicles it has never driven before)

1. Any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks (Stanford HAI)

Generative AI is well suited to several applications but not well suited for others



Applications suited for GenAI

Generating content, possibly while taking context or instructions into account

- Personalized image/audio/video generation and editing
- Chat-like customer service applications

Predicting or extracting information from unstructured data

- Synthesis, question-answering, reasoning about text, images, and multi-modal data
- Search, topic/key information extraction



Applications NOT suited for GenAI

High-stakes scenarios where errors, factual inaccuracies, or value judgements can cause harm

Applications involving **heavy volume of requests and/or tight response time limits** (powerful generative models are comparatively costly and slow)

Unconstrained, long, open-ended generation that may expose harmful or biased content to users

Applications requiring **explainability** and/or **full understanding of potential failure** modes (e.g., highly regulated environments)

Applications requiring **numerical reasoning** (from basic arithmetic to optimization)¹

1. Current topic of research: how to use GPT-like models to generate code that involves solving numerical problems

Four main productivity use cases are already starting to prove significant value



Coding

Interpret, translate, and generate code (e.g., migration from legacy systems at scale, automated development of tests, documentation, and linting)

Generate synthetic data
Prototype applications



>55%

Productivity gains for developers utilizing coding co-pilots such as Github Copilot



Customer engagement

Use co-pilots to guide customers through their personalized journeys

Provide 24/7 customer service through intelligent chatbots

Pursue continuous product improvements from customer insights



>60%

AI-driven automation potential of customer interaction volumes over next 5-10 years



Content generation

Create content (e.g., contracts, NDAs, etc.) to reduce manual work

Create personalized communication, design, and next-product-to-buy recommendations (e.g., outbound marketing, emails, website content)



~80%

User adoption of the AI legal assistant Harvey, with usage of at least once per month



Concision (Virtual Expert)

Summarize and extract insights from unstructured data sources

Retrieve information efficiently

Identify and validate sources of information to enhance credibility

Assist with problem-solving

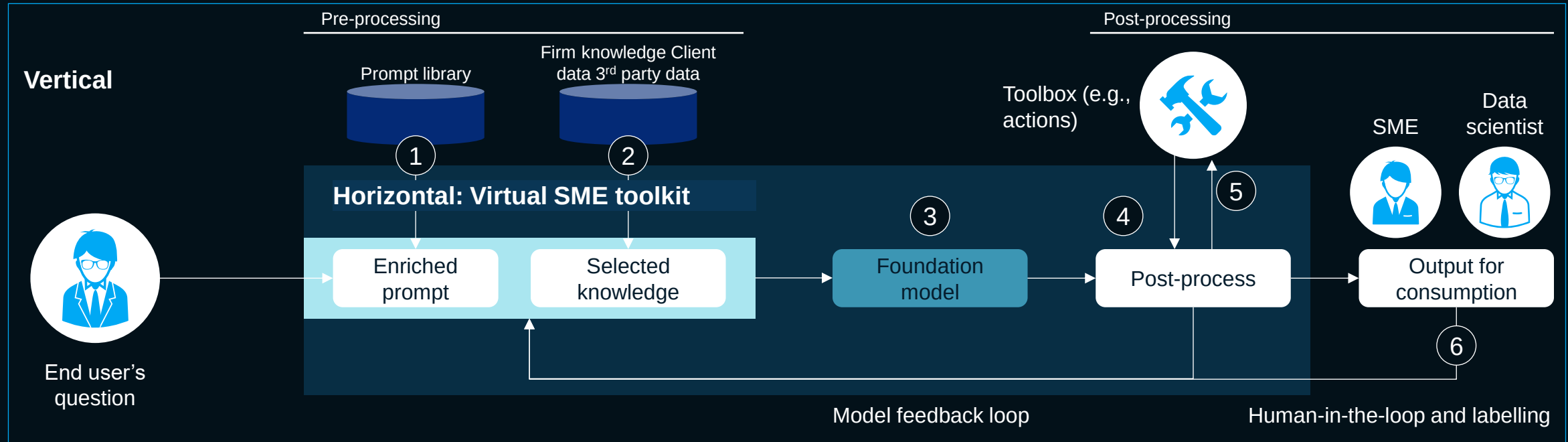


~80%

Productivity gains in verifying statements in news and social media

GenAI orchestration is significantly more than just "calling an API"

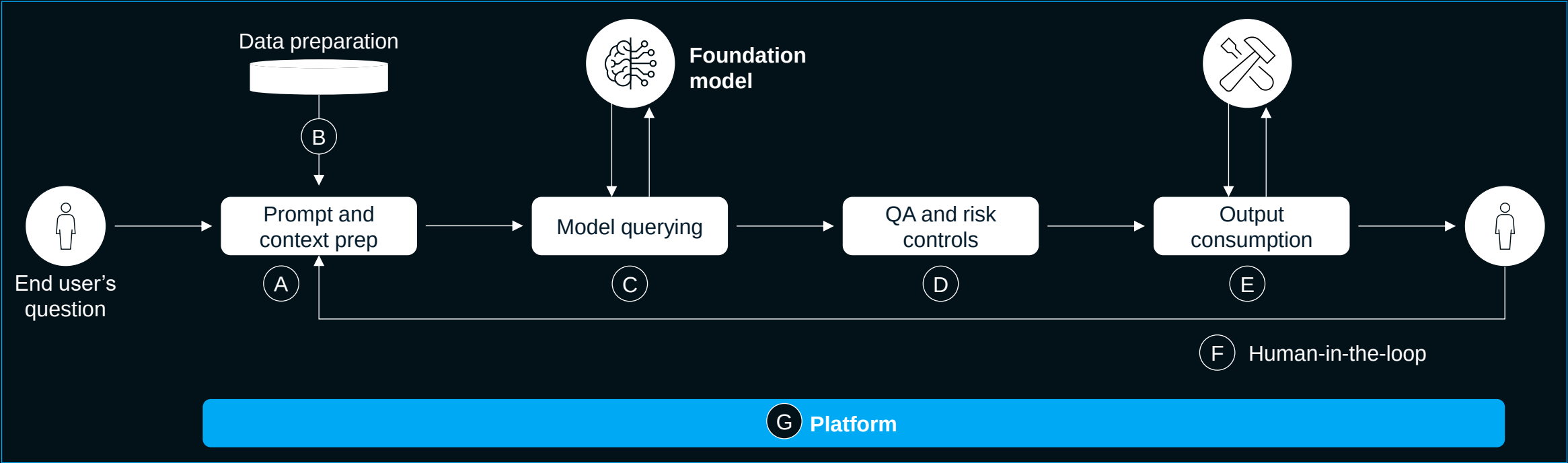
Example: Virtual SME solution architecture



Horizontal toolkit key components

- 1 Prompt engineering:** How do I instruct the model to maximize the odds I get an accurate output?
- 2 Context engineering:** What is the most relevant information to provide the model given my prompt?
- 3 Foundation model:** How can I tailor the model based on my specific domain and task?
Models have captured the most attention, but these will drive limited impact in isolation
- 4 Post-processing:** How do I validate the model's output to ensure its accurate, compliant, and fair (e.g., source attribution)?
- 5 Toolbox:** How do I give the large language model additional capabilities to get the desired end result?
- 6 Human-in-the-loop:** How can I work with SMEs to ensure accuracy and enable a learning flywheel for scalability?

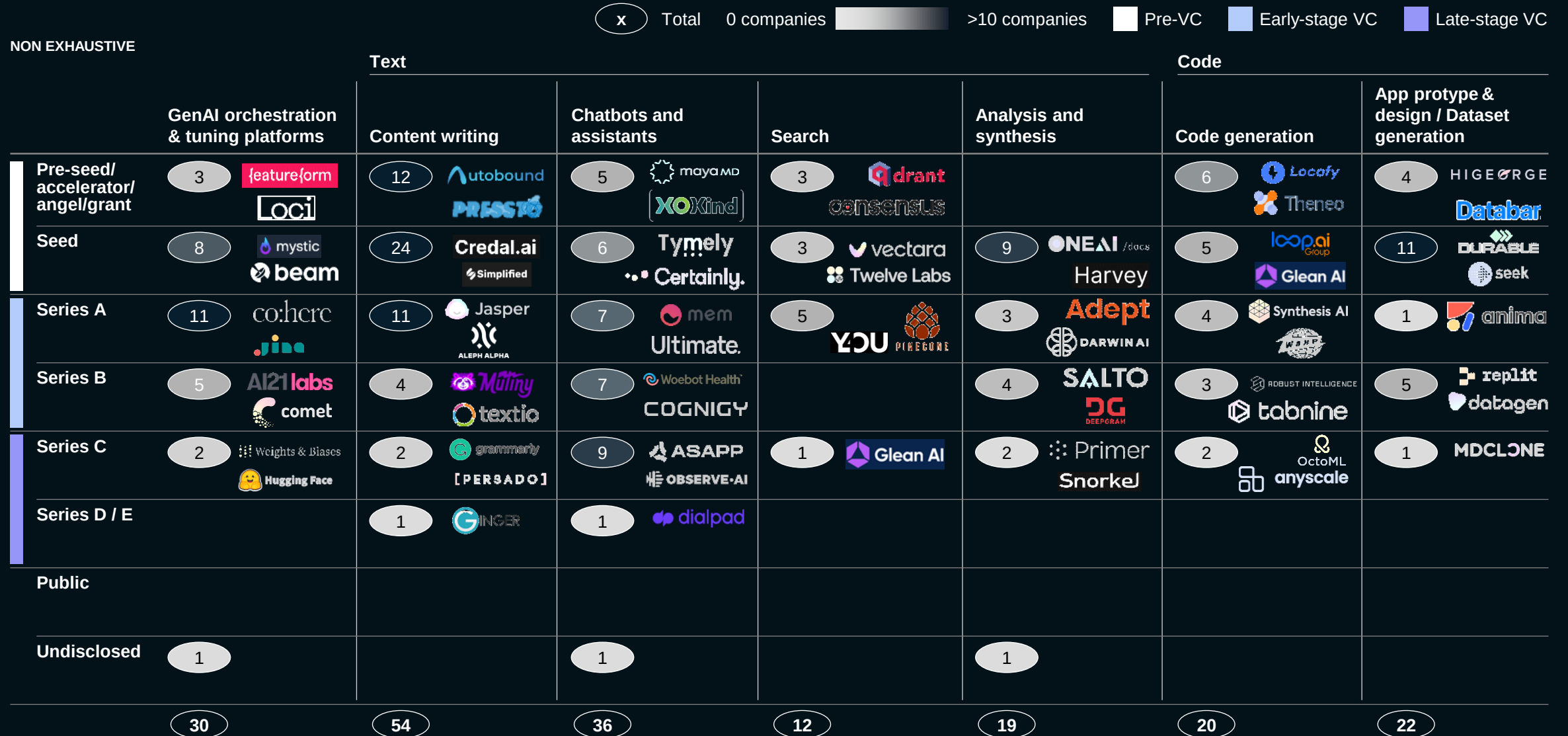
Example of tools and playbooks we have developed to enable rapid and high quality deployment



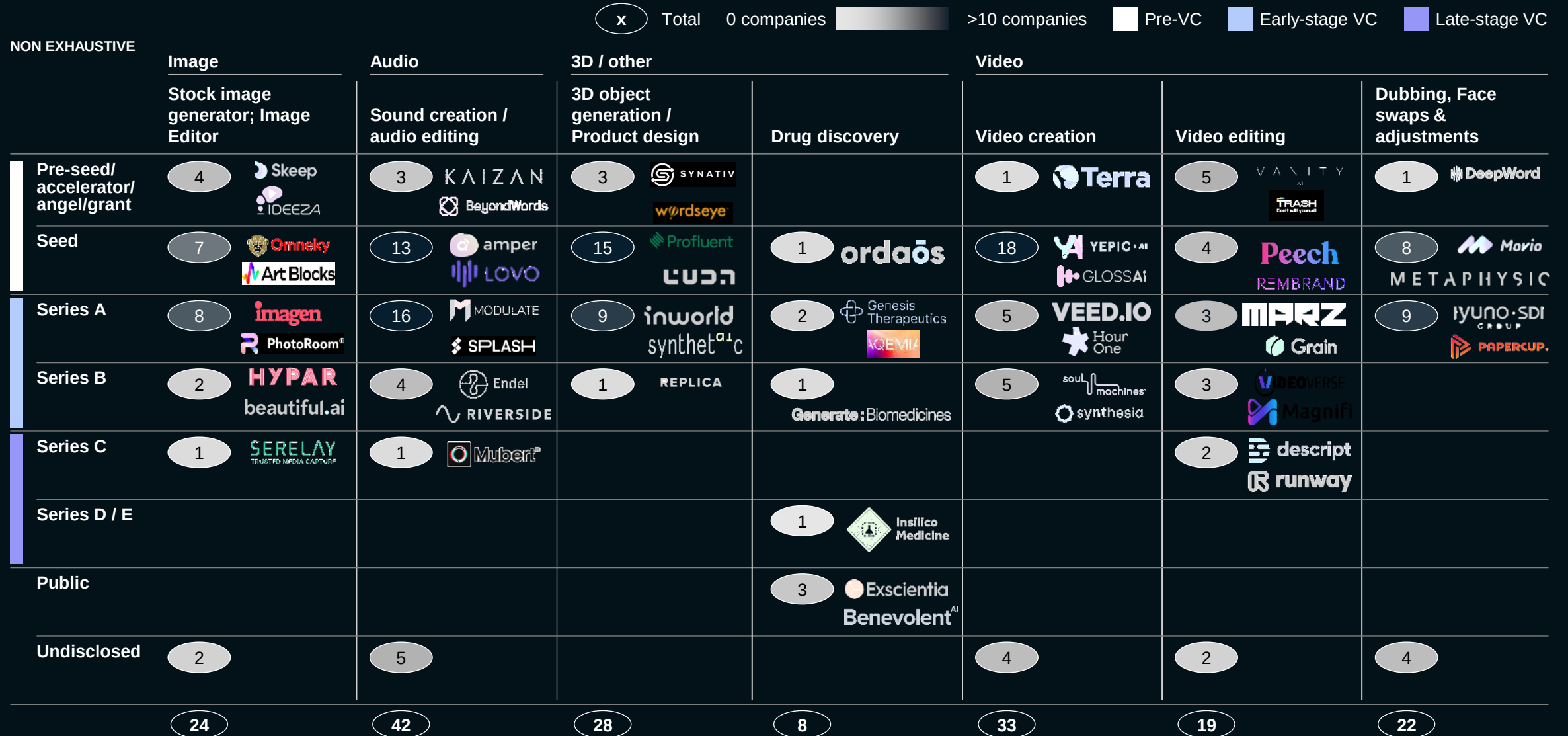
Key components

(A)	(B)	(C)	(D)	(E)	(F)	(G)
Prompt library	Vector embeddings	Model selection	Self-consistency	Action triggers (APIs)	Human feedback	IaC blueprints
In-context learning	Semantic search	Parameter tweaking	Source attribution	Visualization tools	Reinforcement learning HF	API gateway
Chain of thought / Virtual Agent	Knowledge graphs	Memory	Model confidence			Model hosting and training
		Fine-tuning	Toxicity score			

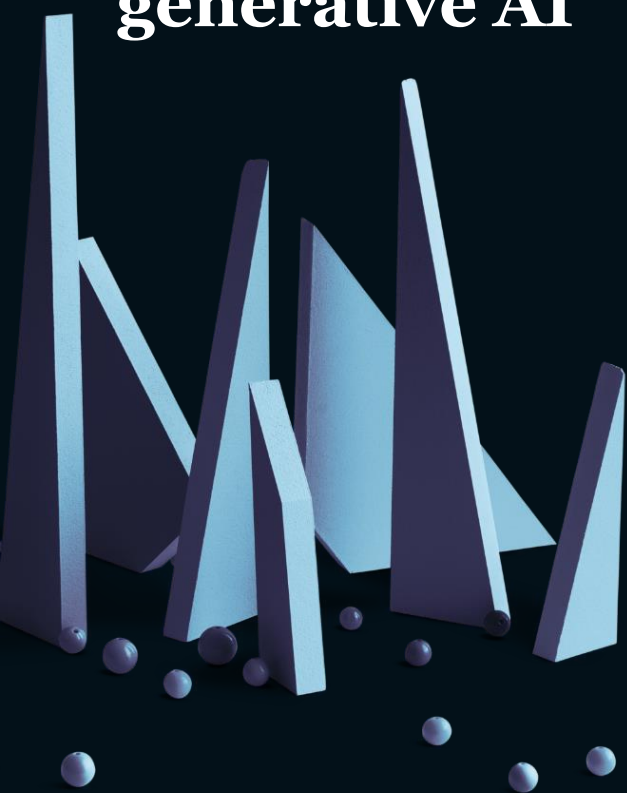
End-user applications most mature in text modality, as seen by later stage VC funding rounds (1/2)



End-user applications most mature in text modality, as seen by later stage VC funding rounds (2/2)



There are 8 key risk categories associated with the use of generative AI



Risk category	Description of potential risk	Impact of generative AI on risk landscape
Impaired fairness	Algorithmic bias; misrepresentation of generated content as human-created	Generative AI can produce content that is not clearly identifiable as AI-generated, leading to confusion or deception of users
IP infringement	Infringement on copyrighted or otherwise legally protected materials	Foundation models typically leverage internet-based data, leading to incidents of IP infringement (e.g., copyright violations, plagiarism)
Privacy concerns	Unauthorized use/disclosure of personal or sensitive information	Generative AI may heighten privacy concerns through use of personal or sensitive information for model training
Malicious use	AI-generated promulgation of malicious content	Generative AI, particularly open-source LLMs, can be leveraged to create and disseminate malicious content (e.g., falsehoods)
Security threats	Vulnerabilities in generative AI systems that may be breached or exploited	Generative AI applications may be subject to prompt injection and other vulnerabilities (e.g., payload splitting to bypass safety filters)
Performance & explainability risk	Inability to explain model outputs appropriately and model inaccuracies	Foundation models may generate factually incorrect or outdated answers (e.g., hallucination)
ESG impact	Non-compliance with ESG standards; reputational risk	Training and deployment of foundation models may increase carbon emissions and exceed ESG commitments or lead to workforce disruption
Third-party risk	Risks associated with the use of third-party AI tools	Use of third-party generative AI models and tools can impose risk of proprietary data being used by public models

Key risks can further lead to **regulatory, legal, reputational and business consequences**

“Agent systems” appear to be unlocking a new horizon in GenAI accuracy and productivity

What value can Agents bring to businesses?

Unlock potential to automate a long tail of complex use cases characterized by highly variable inputs and outputs or time-efficient manner through



Agents can support high-complexity use cases across industries and business functions, particularly for workflows involving time-consuming tasks or requiring various specialized analysis through 4 steps:

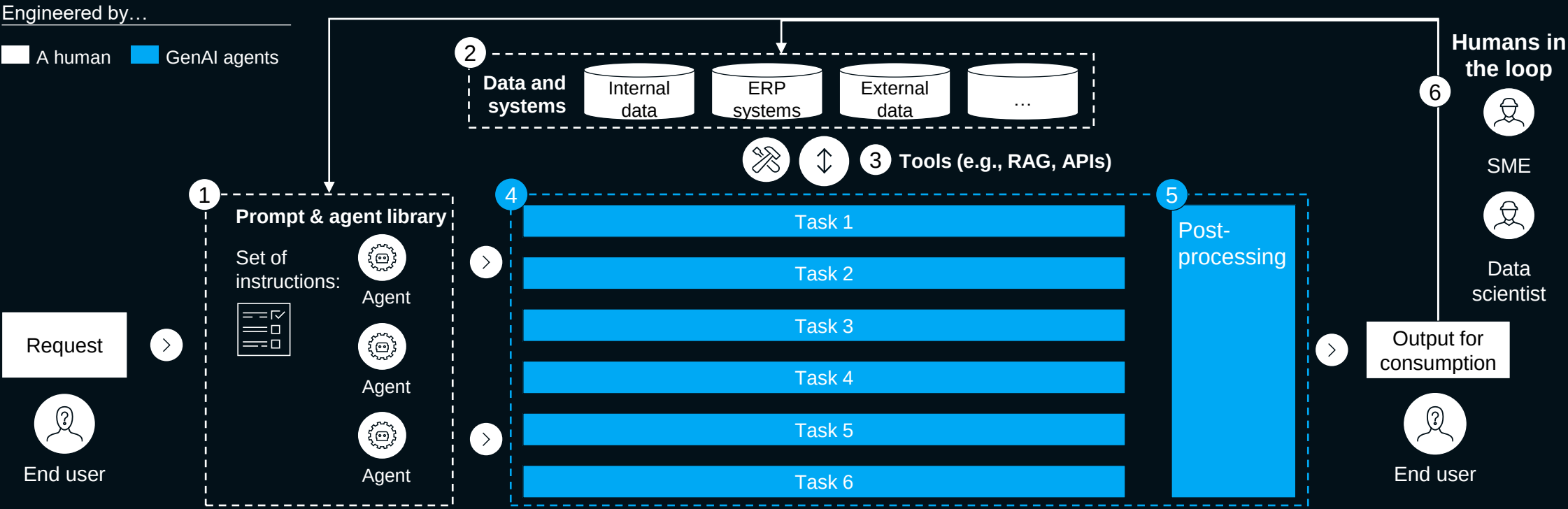
1 Managing multiplicity: potential to handle variety of situations, adapting in real time to perform specialized tasks required

2 Being directed with natural language: using natural language as a form of instruction

3 Working with existing software tools & platforms: using tools and communicate across a broader digital ecosystem



Agent systems can that communicate across multiple tasks can produce more accurate/sophisticated outputs, supported by post-processing and human-in-the-loop



- 1

Prompt & agent library: A set of instructions crafted by your SMEs that enshrine your domain expertise, which agents use for tasks
- 2

Data and systems: How do I organize my data and IT systems to ground AI in my organization's context?
- 3

Tools: How do I give the GenAI agents the ability to interact with the "real world" (e.g., read, write, take actions)?
- 4

Agents: Agents learn from domain experts to self-organize to break down tasks into subtasks and execute them using tools and data
- 5

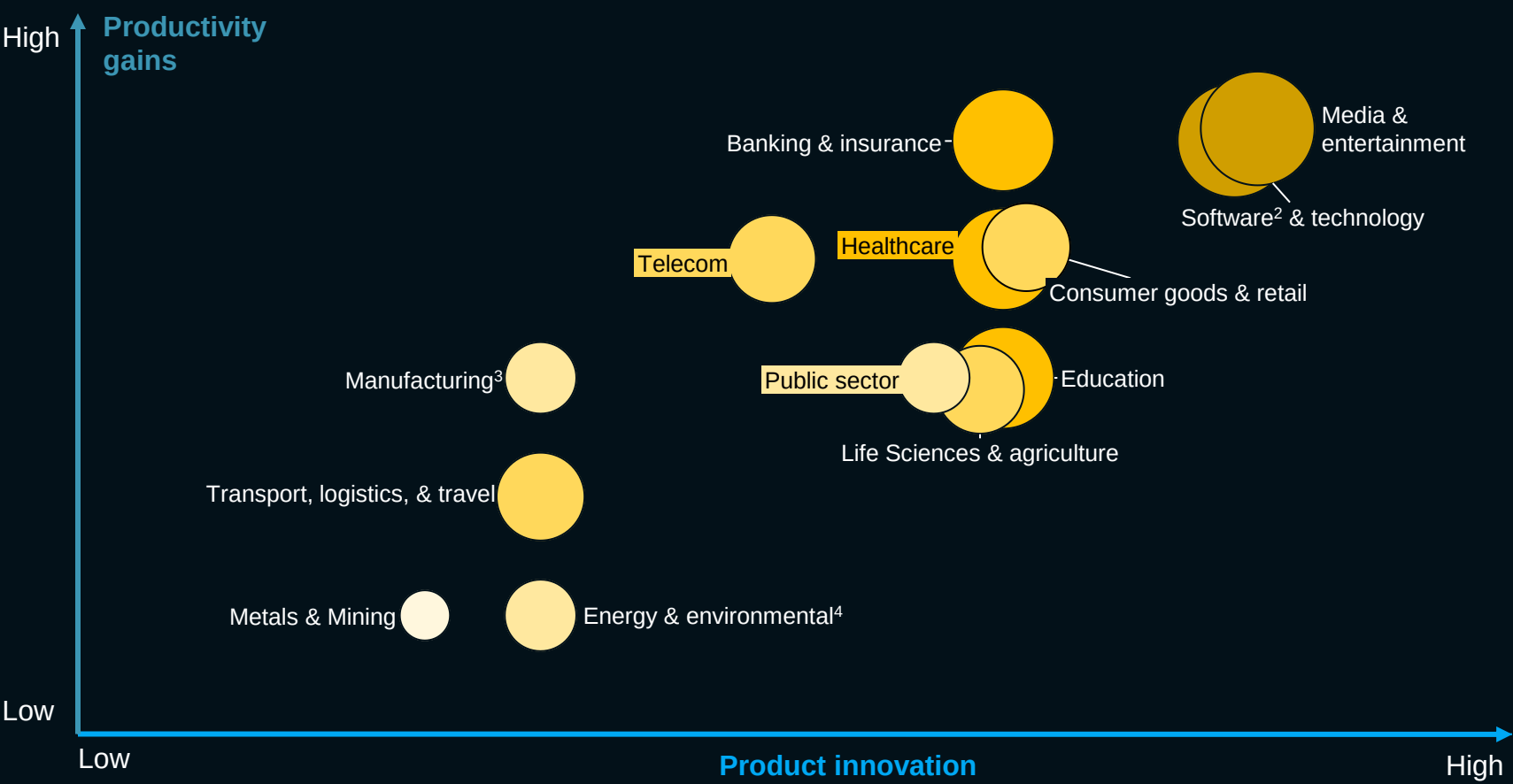
Post-processing: Agents learn from domain experts on "what good looks like" to verify outputs' accuracy, compliance, and fairness
- 6

Human-in-the-loop: How can I work with SMEs to ensure accuracy and enable a learning flywheel for performance, scalability?

Generative AI will impact all sectors, with highest disruption expected in Software & technology, and Media & entertainment



Impact of GenAI across industries¹



Key takeaways

- Media & entertainment** paradigm shift from artificial content generation and media co-pilots enabling novel hyper-personalized content creation at low costs
- Software² & technology** impacted by radical developer productivity growth and new conversational interaction patterns across solutions
- Banking & insurance** impacted by massive automation potential and personalization opportunities

Agenda

The potential of Generative AI

Demos – Virtual Agents and LegacyX

We conducted an exploration effort with a leading US bank to automate credit memo drafting using generative AI (GenAI)

Writing credit memos is the client's first use case of GenAI on their transformation journey



Challenges

Time-consuming: Relationship Managers (RMs) and Credit Analysts (CAs) spend 2-4 days writing each credit risk memo to inform corporate banks' credit decisions or fulfill regulatory requirements

Numerous data sources: CAs manually review and extract info from 10+ data sources (e.g., credit rating reports, broker reports, financial databases, news, meeting notes)

Complex dependencies & reasoning: Credit memos can be 10-20 pages long, requiring nuanced reasoning across interdependent sections

No room for error: Credit analysis needs to be frequently updated, carefully sourced, double-checked, and syndicated across divisions

Low client readiness: Writing credit memos is the client's first GenAI use case and their technology readiness is low



Our approach

Reimagined their end-to-end credit risk process with GenAI and produced a vision prototype for the board

Conducted user research (e.g., user interviews, concept testing) with Credit Analyst teams to understand their processes and pain points

Prioritized credit memo writing in the credit risk domain due to the high manual effort, a large amount of qualitative data, and higher feasibility of building GenAI solutions

Focused on automating writing the financial analysis section to uncover higher-order insights into borrowers' profitability, liquidity, and solvency

Designed technology architecture, laid out the key design choices, and hosted a day-long workshop with 30+ clients

Defined the operating model and roles of the joint client-McKinsey team, as well as a detailed roadmap for delivery



Projected impact

20-60%

CA productivity gain

~30%

Faster decisioning speed



The approach at 50,000 feet

Example for legacy risk model modernization



Legacy code
(e.g., SAS)

Traditional automated approach for legacy modernization

One-step direct conversion

of legacy code into modern code preserving code structure and organization



Modern code
(e.g., Python)

Key takeaways

Resulting code is 1:1 functionality-wise and typically of low quality - still requires significant effort to modernize end-to-end



Legacy code
(e.g., SAS)

GenAI-enabled legacy modernization approach

Gen-AI accelerated migration is up to 50% faster

Reverse engineer

Discover risk model code, algorithms & pipelines
Extract insights via GenAI, e.g., human readable model docs, bus. rules
Generate test cases to be used to check new code

Shape

Define future state vision & architecture
Create roadmap and wave plan for execution
Refine autogenerated model specification to reflect vision

Modernize

Convert refined model specification to modern language
Test, refine and deploy new code to target environment
Performance-test new code vs. legacy



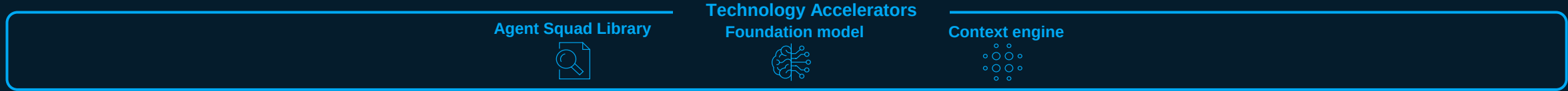
Modern code
(e.g., Python)

New functionality can be seamlessly added in the Shape stage by adjusting model specification

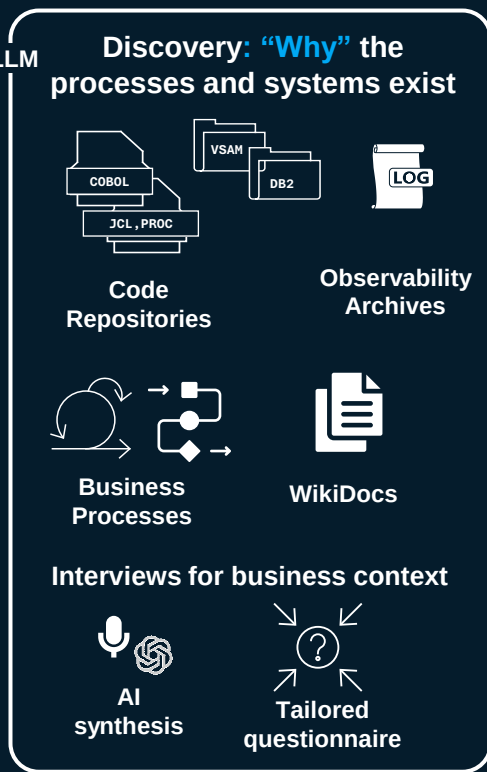
Resulting code of higher quality as no direct conversion and anchoring to old code structures

Thus, significant acceleration can be seen

The approach derives intent of legacy systems to accelerate modernizations using GenAI supported by developers in the loop

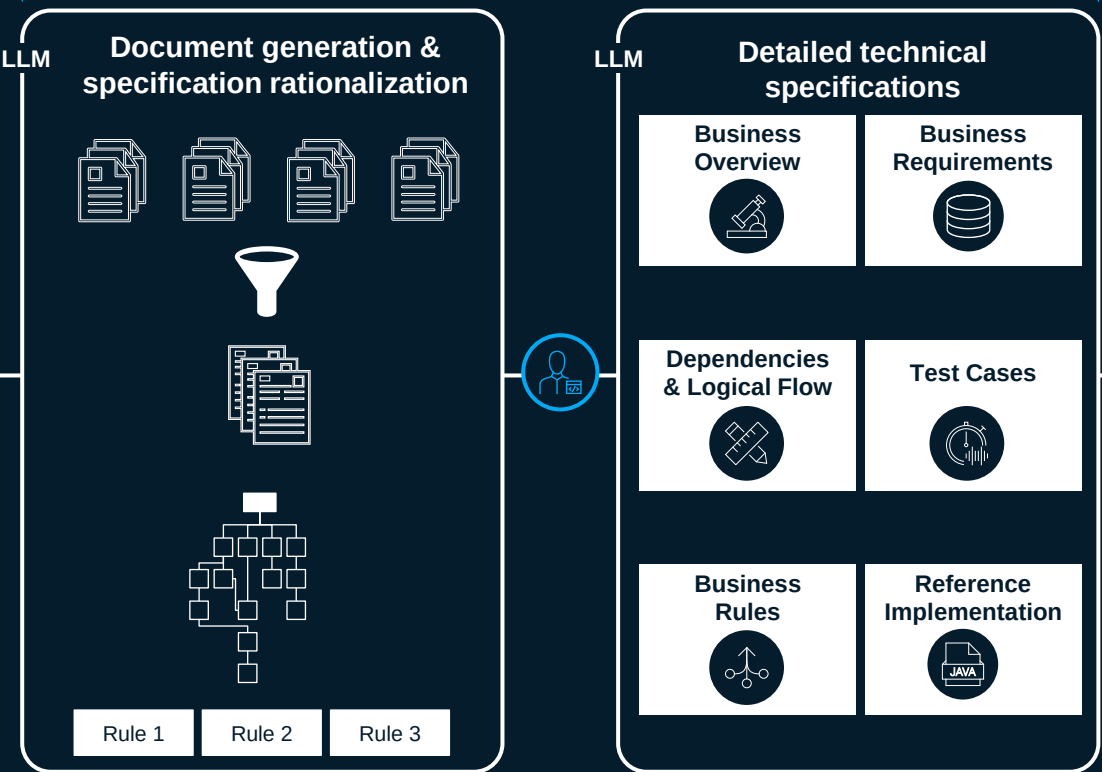


Step 1: Reverse-engineer



Cross-bounded context and data inputs including subject matter expert interviews, logs, WikiDocs to gather business context, process insight, code logic, etc.

Step 2: Shape



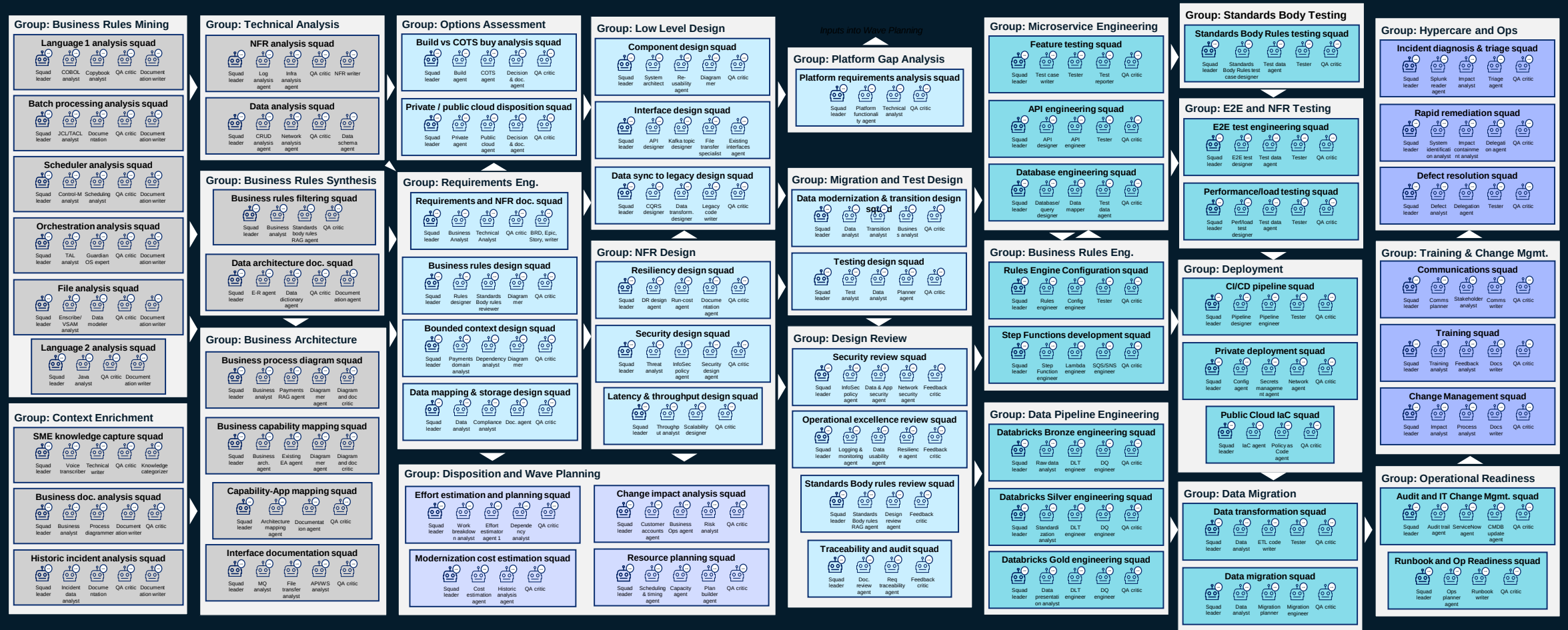
Rationalization of specifications based on documentation review (e.g., combining nested procedural logic, simplification of data structures, etc.)

Detailed technical design based on reference implementation and including unit test cases

Generation of converted code and data with self-validation loops and checks to ensure all business logic has been captured

A tiered multi-agent factory framework enables automation of complex software development flows

ILLUSTRATIVE & NOT EXHAUSTIVE





QuantumBlack
AI by McKinsey

Questions? Please reach out . . .



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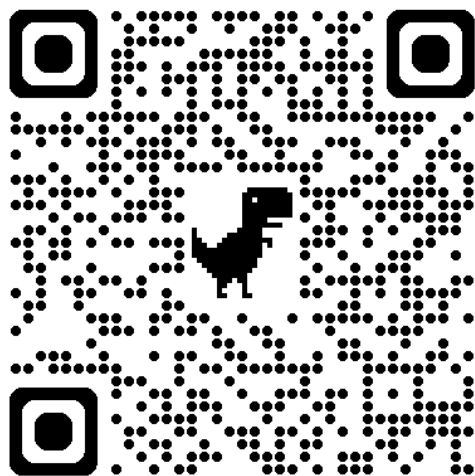


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